

Business Requirements Document (BRD) – PhorSense (AI-Enhanced Smart Payment Routing & Instant Retry Mechanism)

*self-directed PM/BA simulation exercise, modeled on a mid-size Indian payments aggregator's
orchestration platform (Nimbus Transaction Router)*

NimbusPay Technologies Pvt. Ltd.
Pune, India (Simulated)

PhorSense – AI-Enhanced Smart Payment Routing & Instant Retry Mechanism

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<Position: Project Manager / Business Analyst (Portfolio Exercise)>

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Document History				
Revision	Approved By	Revision Date	Description	Author
v1.0	Self- Reviewed	20/09/2026	Initial BRD drafted for PhorSense routing & retry initiative	Manik Tangade
v1.1	Self- Reviewed	22/09/2026	Added functional requirements, risk analysis, and cost-benefit sections	Manik Tangade

Project Brief

Executive Summary

This BRD defines the business needs, scope, and functional requirements for PhorSense, an AI-enhanced payment orchestration layer that sits on top of NimbusPay's existing rule-based Nimbus Transaction Router (NTR).

The objective is to improve transaction success rates, reduce false declines, and minimize revenue loss from failed payments by introducing real-time ML-based routing, compliance-aware instant retries, and an integrated risk-scoring engine without disrupting the existing routing infrastructure merchants already depend on.

This BRD is intended for project stakeholders, including Product Owners, Project Managers, Developers, QA teams, and PMO leadership, to ensure alignment on delivery goals, responsibilities, and success criteria.

Project Description

The PhorSense project supports end-to-end delivery of an AI routing/retry layer through structured sprint planning, Jira tracking, weekly PMO reporting, and risk management.

Currently, routing and retry decisions are static and rule-based, and risk, compliance, and delivery updates are tracked inconsistently, reducing visibility and delaying decisions. This project addresses that through standardized coordination practices.

Project Drivers

Driver	Description	Desired Output
Success-Rate Visibility	Static routing gives limited insight into real-time route performance	Centralized ML-scored routing with live tracking
Process Efficiency	Manual, inconsistent status reporting across teams	Standard weekly status reports
Risk & Compliance	Retry/compliance risks identified late in the delivery cycle	Proactive RAID tracking, gated compliance sign-off

Project Description

This project focuses on coordinating the Agile delivery of PhorSense through structured planning, tracking, reporting, and risk management activities.

Objectives

- Improve success-rate visibility and delivery health tracking
- Ensure timely coordination between Data Science, Engineering, and Risk & Compliance
- Proactively identify and manage risks, issues, and dependencies
- Provide clear, stakeholder-ready status reporting

Tasks

- Coordinate sprint planning and execution activities
- Track features, tasks, and defects in Jira
- Maintain RAID logs (Risks, Assumptions, Issues, Dependencies)
- Prepare and circulate weekly project status reports
- Support UAT coordination and defect tracking
- Support shadow-mode/pilot testing and defect tracking

Deliverables

- Weekly Project Status Report (Excel)
- RAID Log (Risks, Assumptions, Issues, Dependencies)
- Jira Board with sprint tasks, features, and defects
- UAT defect tracking and closure reports
- Stakeholder presentations and MoM documentation

Costs

- Internal/self-directed simulation project; no external budget tracked
- Effort costs limited to team coordination and reporting activities
- Tool usage: Jira, Excel, PowerPoint (no additional licensing cost)

Deadlines

- Sprint duration: 2 weeks
- Weekly status reporting cadence
- Compliance sign-off gated before pilot cutover
- Target: phased pilot within ~5 months of kickoff

Project Requirements

Functional Requirements

Requirement	Description	Priority
Smart Routing	Rank gateway/acquirer routes in real time by success probability (Availability → Performance → Cost → Profit)	High
Smart Retry	Classify declines as soft/hard and gate every retry through compliance and attempt-cap checks	High
Risk Engine gating	Score every transaction (Allow/Block/Step-up) before routing; a Block is final	High
Sprint & workflow tracking	Track sprint tasks across To Do → In Progress → Review/Testing → Done	Mid

Non-Functional Requirements

Requirement	Description	Priority
Latency	Routing/risk decisions must stay imperceptible to checkout UX	High
Compliance	PCI DSS v4.0.1, RBI PA Master Direction, and network retry rules always met	High
Data consistency	Task, risk, and milestone data stays consistent across Jira, RAID log, and reports	Mid
Usability	Dashboards/SOPs are usable by cross-functional teams with minimal training	Mid

Project Factors

Stakeholders

Name	Position	Role
Priya Nair (Simulated)	Product Lead	Business rules, merchant priorities, roadmap
Arjun Mehta (Simulated)	Engineering Lead	Routing/Retry engine build, API integration
Kavya Reddy (Simulated)	Risk & Compliance Lead	Decline taxonomy, PCI/RBI/CERT-In sign-off

Budget

Item	Description	Budget (₹)
Project Team	PM, Dev Team, Scrum Master coordination	28,00,000
ML Infrastructure	Feature store, training/serving, shadow-mode environment	1,50,000
Tools & Reporting	Jira, Excel, dashboard/reporting utilities	30,000
Misc Fees	Meetings, documentation, coordination overhead	40,000
Total Budget		30,20,000

Internal simulation: costs are indicative, for planning/reporting practice only.

Resources

Resources	Qty	Unit Cost (₹)
Project Manager	1	4,00,000
Development Team	6	3,50,000
Scrum Master	1	3,00,000
Jira & Reporting Tools	1 set	30,000

8-person Scrum team over 5 months. There are no separate QA/DevOps/Data Science sub-teams. All are folded into the cross-functional Development Team.

Schedule

Milestone	Completion Date	Responsible
Discovery & scope sign-off	04-Sep-2026	Project Manager
Data Foundation complete	18-Sep-2026	Development Team
Compliance Review (PCI/RBI/CERT-In)	25-Sep-2026	Project Manager
Model Development complete	09-Oct-2026	Development Team
Engine Build complete	05-Nov-2026	Development Team
Shadow/Canary Testing complete	03-Dec-2026	Development Team
Pilots go/no-go decision	04-Jan-2027	Project Manager
Scaled Rollout complete	25-Jan-2027	Development Team

Risk Analysis

Risk	Impact Level	Action Plan
1. Retrying against issuer "do-not-retry" signals triggers network penalties	High	Strict soft/hard classification compliance gate before any retry
2. Model underperforms without sufficient historical data	Mid	Extend shadow-mode observation; use continuous documented evidence
3. Smart Retry incorrectly overrides a Risk Engine Block	High	Block hard coded as terminal validated in shadow mode before pilot
4. 6-person team scheduled across parallel workstreams (model tracks, 3-way engine build)	Mid	Document informal sub-pairing; monitor context-switching load

Cost & Benefit analysis

Cost	Benefit
Team time for sprint planning, coordination, and reporting	Improved cross-functional collaboration (Product, Engineering, Data Science, Compliance)
ML infrastructure and tooling (Jira, dashboards)	Increased delivery efficiency through sprint tracking and PMO governance
Compliance review effort	Better visibility into delivery health via status reports and RAID logs
	Reduced revenue leakage from failed payments

Projected Cost: ₹30,20,000

Projected ROI (Year 1, indicative): ₹75,00,000